Name: Class: Date:

Activity 1 · Hands-on · 🌳 Outdoors · ⏱ 1 hour

🔎 Campus Plant-Group Hunt

You will learn: Plants climb a ladder of complexity — algae, bryophytes, pteridophytes, gymnosperms and angiosperms.

YOU NEED

- Notebook and pencil
- Phone camera
- A hand lens (optional)

⚠️ **Safety:** Do not pluck rare plants. Stay away from deep water and thorny bushes. Wash hands after touching wet walls.

STEPS

1. Look for green slimy patches on wet walls, ponds or water tanks — algae.
2. Look for soft green carpets on damp walls or tree bark — mosses (bryophytes).
3. Find a fern in a shady, moist place — look under the leaves for brown dots.
4. Find a gymnosperm in a garden — Cycas, Thuja, Araucaria (Christmas tree) or Pinus.
5. Find any flowering plant with fruits — an angiosperm.
6. Photograph each one and fill in the table.

OBSERVATION TABLE

Plant found	Where it grew	Group	Vascular tissue?	Seeds?	Flowers / fruits?

QUESTIONS

1. Why do mosses grow only in damp places?

2. Which group was the first to have xylem and phloem?

3. How is a gymnosperm seed different from an angiosperm seed?

WHAT IS HAPPENING? — THE SCIENCE

Algae have no roots, stems or leaves. Bryophytes live on land but need water for fertilisation and have no vascular tissue. Pteridophytes are the first land plants with xylem and phloem but have no seeds. Gymnosperms have naked seeds. Angiosperms have flowers and seeds inside fruits.

Key terms: algae · bryophytes · pteridophytes · gymnosperms · angiosperms

Activity 2 · Hands-on · 🏠 At home · ⏱ 1 night

🌿 Fern Spore Print

You will learn: The fern plant you see is the sporophyte; it makes spores in sori on the underside of its leaves.

YOU NEED

- A mature fern leaf with brown dots underneath
- A sheet of white paper
- A bowl to cover the leaf
- Phone microscope or hand lens

⚠️ **Safety:** Take only one leaf from a big plant. Wash your hands after handling spores.

STEPS

1. Look at the underside of the fern leaf — the brown dots are sori.
2. Observe a sorus with the phone microscope and sketch it.
3. Place the leaf, underside down, on white paper.
4. Cover it with a bowl and leave it overnight.
5. Next morning lift the leaf gently and observe the brown dust pattern (spore print).
6. Look at the spores at high zoom and describe their shape.

OBSERVATION TABLE

Observation	What I saw

QUESTIONS

1. Is the fern plant a sporophyte or a gametophyte?

2. What is a sorus?

3. What grows from a fern spore?

WHAT IS HAPPENING? — THE SCIENCE

In ferns the main plant body is the diploid sporophyte. Sporangia grouped in sori release haploid spores, which grow into a small heart-shaped gametophyte (prothallus). This shows the dominant sporophyte of pteridophytes.

Key terms: **sporophyte** **sori** **sporangium** **prothallus**

Digital activity

💻 Climb the Plant Ladder

> Tap a plant, then tap its group.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch3/2**

ALGAE · BRYOPHYTES · PTERIDOPHYTES · GYMNOSPERMS · ANGIOSPERMS

Algae	Bryophytes	Pteridophytes	Gymnosperms	Angiosperms

Spirogyra · Chlamydomonas · Laminaria · Polysiphonia · Marchantia · Funaria · Sphagnum · Selaginella · Equisetum · Dryopteris · Cycas · Pinus · Ginkgo · Wolffia · Mango · Sunflower

> **You will learn:** Algae → Bryophytes → Pteridophytes → Gymnosperms → Angiosperms: vascular tissue, seeds and then fruits appear step by step.

THINK & WRITE

1. Which two groups have no vascular tissue?

2. Which group has seeds but no fruits?